

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DRK-010 - Dirok-010

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000116
Field / District	Dirok / Dibrugarh
Block	OALP-III-08
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2017-09-23 / 2018-01-31
Measured / TVD	2160 m / 2116 m
Trajectory	Deviated; inc 27.7 deg; az 299.1 deg
Formation Interval	1357-2051 m
Lithology	Fine-medium sandstone with shale interbeds
Pore Pressure Reference	2562 psi
Fracture Gradient	15.26 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Moderate (45.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	540	Surface
9 5/8 in casing	1252	Intermediate
7 in liner/casing	1900	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2017-11-29 1214 m	Mud Loss	Critical	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2017-12-21 1552 m	Mud Loss	Critical	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.
2018-01-11 1867 m	Mud Loss	Critical	Active pit volume declined during stable pump rate. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2017-10-21 591 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2017-10-31 749 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Condition hole and confirm pressure normalization. Lesson: Correlate pressure anomalies with flow before escalating risk.
2017-12-11 1389 m	Kick	High	Drilling break followed by positive flow check. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2017-10-25 656 m	Torque Spike	Medium	Torque rose without corresponding pressure anomaly. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.

Completion Summary

The synthetic well reached 2160 m MD in the Tipam interval. The NWIS knowledge record links 14 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.